



A single-chip Split-6 PHY implementation for 5G NR, using RFSoc and PYNQ

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ABSTRACT

Indoor traffic has been estimated to account for over 70% of all mobile data in urban areas. We propose a small cell Split-6 radio unit (RU) to meet the indoor use case, presenting a hardware implementation of the physical layer Physical Downlink Shared Channel, targeting an AMD UltraScale+ Radio Frequency System on Chip (RFSoc) device. Several custom IP cores are developed and used in conjunction with integrated resources available on the target RFSoc chip. The design uses less than 10% of Block RAM memory, and under 3% of other available Programmable Logic resources; meeting timing at 122.88 MHz, enabling 100 MHz bandwidth channels. This modest resource utilisation leaves capacity for integration of other physical layer channels. Total power of 4.33 W and dynamic power of 2.533 W confirm that the baseband RFSoc power draw is very low. A test harness developed using the PYNQ software/hardware framework is presented, which provides a low-complexity approach to radio testing without external network software. The PYNQ system provides a user interface to update resource allocation in real time, and produces resource grid and constellation plots based on data captured from the design operating on the target RFSoc4x2 development board. The system is tested through two case studies, validating each sub-step of the physical layer. Latencies of 0.332 ms and 0.719 ms, with a 122.88 MHz clock, meet key HARQ requirements. The success of the study confirms RFSoc devices as candidates for small cell Split-6 deployment.

1. Introduction

The fifth generation of cellular network technology, 5G New Radio (5G NR), is defined by the standards body 3rd Generation Partnership Project (3GPP) from introduction as Release 15 in October 2019 [1], through to Release 18 in April 2025 [2]. Mobile broadband 5G NR has been demonstrated in a wide range of applications from high bandwidth video streaming to minimal latency, timing-critical machine-to-machine communication and Internet of Things environments. A key insight into modern wireless network usage is that the majority of mobile data traffic is generated indoors. Both Ericsson [3] and Nokia [4] report that indoor usage accounts for 70%–80% of all mobile traffic in urban areas. This gives a strong indication that mobile data is utilised in the same manner as a Wi-Fi network, and as such the reliable, high-bandwidth streaming of 5G NR is well suited to this environment. The majority of current 5G NR urban coverage is between 410 MHz to 7.125 GHz, known as Frequency Range 1 (FR1). Traditional outdoor 5G NR base stations often struggle to address this indoor component of user demand, due to the attenuation of radio signals through walls and windows at these multiple GHz frequency bands [5].

One approach to this issue is a small cell network, where compact, low-range, low-power base stations can be deployed to supplement existing Mobile Network Operator (MNO) networks. The work presented in this paper demonstrates the hardware implementation of

the entire physical (PHY) layer of a 5G NR transmitter, i.e. all data processing stages, on the radio portion of a network. This is strongly motivated by the deployment of small cells, where a low number of radios can be deployed to provide coverage over a specific area. Small cell deployment is driven by two factors: the addition of coverage in urban areas to supplement mobile coverage from MNOs, and to provide temporary coverage areas for short term events, such as festivals or sporting events.

The specific motivation for this work is therefore to produce a radio which is well suited to, and improves upon pre-existing, small cell deployment. By implementing the PHY layer data processing on the radio, the network tasks of a traditional base station can be greatly reduced. As such, this work aims to be the first step towards a 5G NR radio which acts similar to a Wi-Fi access point, installed in an indoor environment such as an office or house, which is connected via fibre-optic cable to a centralised base station with lowered requirements compared to a traditional approach.

1.1. Disaggregated radio access networks

Within 5G NR, small cells are typically designed following Open Radio Access Network (O-RAN) methodology as defined by the O-RAN Alliance [6]. O-RAN adopts the concept of a disaggregated radio,

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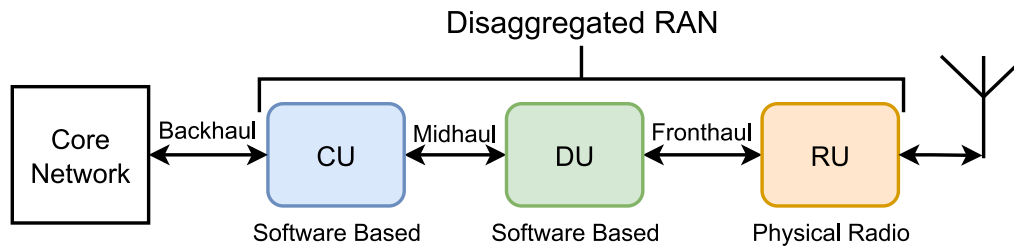


Fig. 1. Overview of a standard network configuration with a core network connected to a disaggregated RAN.

where the functionality of a network can be partitioned between three elements: a Centralised Unit (CU), a Distributed Unit (DU) and a Radio Unit (RU). The three elements of a disaggregated network may collectively be referred to as a gNodeB (gNB). The disaggregation of network components enables considerable flexibility in the deployment of resources depending on traffic needs, geography, and use cases. This disaggregation also reduces the cost and complexity of deploying full baseband processing at each site, perhaps only a single RU would need to be physically deployed for example.

1.1.1. gNB components

Typically, the CU handles network layer tasks such as Radio Resource Control (RRC), which manages the connection between a User Equipment (UE) and the base station, as well as tasks such as Packet Data Convergence Protocol (PDCP), which supports Internet Protocol packet compression and encryption. The DU acts as a middle stage between the network layer and the radio. This includes functionality such as the Radio Link Control (RLC) layer, which aids in data packet segmentation and reassembly; MAC layer for resource allocation and control signalling; as well as the Physical (PHY) layer, which converts data into radio signals. An RU in such a system is simplified, containing the Radio Frequency (RF) capable Digital-to-Analogue (DAC) and Analogue-to-Digital (ADC) converters required for transmission and reception respectively alongside analogue components such as antennas and gain control. Fig. 1 shows a standard network configuration where a core network is connected to a disaggregated RAN. The backhaul connection shown here may be several kilometres long between the core network and CU. The CU can be connected to one or more DUs, often located closer together, via midhaul cables. In turn, each DU can be connected to a number of RUs, likely atop radio masts.

1.1.2. Fronthaul connection

Disaggregated radio is reliant on fibre-optic cabling between DU and RU for front-haul. This has considerably lower power consumption and enables several kilometres between gNB components compared to traditional copper cable. Introduced in 4G Long Term Evolution (4G LTE) and commonly used in 5G NR small cells, fronthaul fibre-optic is enabled by the Common Public Radio Interface (CPRI), which transmits serial frames of complex data and control signals [7]. Due to the point-to-point nature of the CPRI interface it becomes difficult to scale to large network sizes.

5G NR further enables an additional, standardised interface, the enhanced Common Public Radio Interface (eCPRI) [8] which is a more efficient version of the original CPRI standard. Where CPRI is a point-to-point serial interface, eCPRI is an Ethernet-packet based interface and as such transports only active user data packets over fibre cable. This is more bandwidth efficient compared to CPRI which transports samples at all times. This Ethernet nature makes eCPRI flexible, enabling a network to efficiently scale for Multiple-Input Multiple-Output (MIMO) use cases compared to the rigid CPRI.

1.1.3. Virtualised Radio Access Network

As shown through Fig. 2, in a small cell the entire CU/DU stack can be virtualised in software within a server or similar. The core network may reside in the same computing resource as in the Fig. 2 example, or within a separate dedicated device, connected to the CU via backhaul fibre-optic cable. Known as a Virtualised Radio Access Network (vRAN) [9], this software approach enables CU/DU stacks such as the srsRAN [10] or OpenAirInterface [11] projects to be deployed alongside a proprietary core network. The Small Cell Forum (SCF), a strong supporter of small cell deployments, stated in 2021 that trends will shift towards disaggregated vRAN architectures with cloud-based CU/DU and separate RUs [12], with the small cell market valued at \$5.26 billion in 2025 [13]. This prediction has materialised with cloud based vRAN CU/DU options from major vendors such as Samsung [14] and Intel, the latter working alongside HPE [15] and QCT [16].

Small cell networks are also a clear area of interest for MNOs, such as Vodafone, who demonstrated the *LimeNET Micro 2.0* [17] network-in-a-box at Mobile World Conference 2024 [18]. Similarly US-based MNOs Verizon, T-Mobile and AT&T have signed deals to deploy several thousand indoor small cell networks [19].

1.2. Functional splits

The flexibility at the heart of 5G NR extends to how the functionality of the network is partitioned between the CU, DU and RU, known as *functional splits* [20]. There are three popular functional splits, presented in Fig. 3, which dictate the distribution of PHY layer processes between the DU and RU: Split-8, Split-7.2 (with minor variations such as Split-7.1), and Split-6.

1.2.1. Split-8

The division of functionality where all RLC, MAC and PHY layer tasks are performed in the DU, referred to as Split-8, is the most common of these functional splits in a traditional 5G NR small cell network, as all CU/DU tasks of the gNB can be contained within vRAN software. The vRAN CU/DU stack could be paired alongside a simple RU with data converters, analogue front end components and antennas, thus simplifying deployment. Split-8 is therefore equivalent to a 4G LTE CPRI approach where the entire PHY layer processing is centralised. As fully processed, Orthogonal Frequency Division Multiple Access (OFDMA) symbols are transmitted over the fronthaul link between DU and RU, a Split-8 approach can be bandwidth intensive e.g. for a 20 MHz channel bandwidth with 2x2 antenna ports the required fronthaul data rate is 2 Gbps [21]. Similarly a 64x64 antenna array at 200 MHz channel bandwidth would require 640 Gbps data rates over a CPRI connection, which is unmanageable over the potential kilometres between DU and RU.

1.2.2. Split-7.2

There are several other notable functional splits which determine the partition of tasks between the CU, DU and RU. Perhaps the most popular of these alternative methods is Split-7.2 [22], where lower PHY layer functionality is shifted to the RU. As such, the bandwidth requirements between DU and RU can be dramatically reduced as

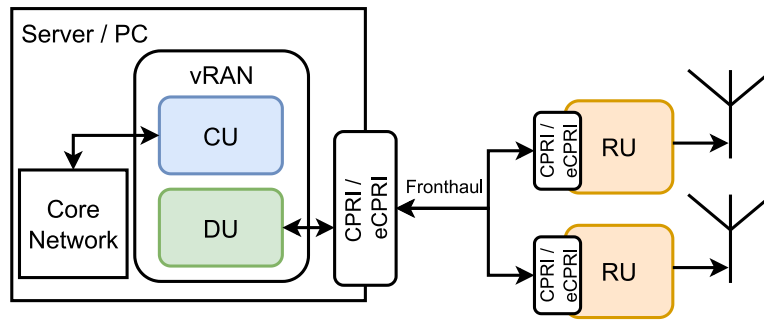


Fig. 2. Overview of a typical small cell network configuration with vRAN.

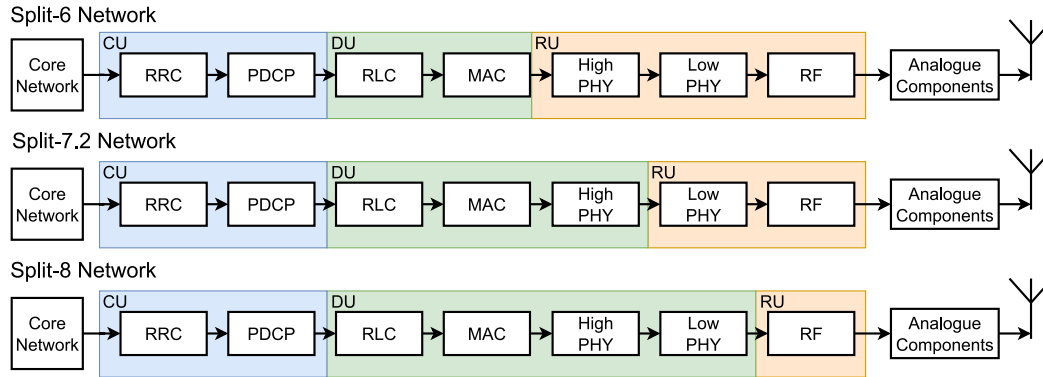


Fig. 3. Block diagram of notable O-RAN functional split options.

Quadrature Amplitude Modulated (QAM) symbols can be compressed on the DU prior to transmission. These symbols can then be decompressed at the RU prior to ODMA symbol generation. Transfers between the DU and RU in a Split-7.2 gNB can utilise the Ethernet-packet based eCPRI standard. For this Split-7.2 approach, bandwidth requirements are around 35% of an equivalent Split-8 CPRI approach [23].

As an Ethernet-based standard, eCPRI, and therefore Split-7.2, is vendor agnostic enabling interoperability between vRAN CU/DU stacks and RUs of different vendors [24]. This allows use of common off-the-shelf (COTS) equipment such as network interface cards and creates an opportunity for smaller, private 5G NR deployments using multi-vendor solutions. However, Split 7.2 networks face a number of key deployment challenges, such as the complexity of matching configurations between gNB and RU. Each gNB vendor and each radio, although having a common interface via eCPRI, may still be non-standard in terms of how they are configured. Therefore, the resulting Split-7.2 system may be far less ‘plug and play’ than an equivalent Split-8 network.

1.2.3. Split-6

An alternative approach to disaggregation exists as Split-6. In such a system, the whole PHY layer is moved to the RU and as such the gNB provides unprocessed data in the form of transport blocks, as well as relevant control packets. As a result, the requirements of the gNB running the CU/DU aspect of the network are greatly reduced, however the demands of each RU are notably increased. Split-6 solutions are primarily championed by the SCF, who define a MAC to PHY layer fronthaul interface, known as network Femtocell Application Programming Interface (nFAP) [25]. The SCF additionally define the theoretical Split-6 network components as the SCF-Radio Unit (S-RU) and a SCF-Distributed Unit (S-DU) [26]. However the testing and implementation of such a system remains scarce, with a lack of vRAN stacks offering S-DU in a similar form as Split-8 and Split-7.2.

1.3. Implementation platform

This work is influenced by Split-8 network-in-a-box designs that feature a virtualised CU/DU and simple RU, and proposes a unique approach to a Split-6 RU deployment. Our approach primarily focuses on the realisation of an S-RU style Split-6 radio through the implementation of the PHY layer on AMD Zynq UltraScale+ Radio Frequency System on Chip (RFSoc) hardware (hereafter referred to as simply ‘RFSoc’).

1.3.1. RFSoc ZU48DR chip

The RFSoc ZU48DR chip, part of the third generation of RFSoc devices [27], is the implementation platform for this design. This chip includes Field Programmable Gate Array (FPGA) Programmable Logic (PL) which will host the PHY layer, constructed from custom Intellectual Property (IP) blocks. The PL is located alongside an Arm-based Processing System (PS) featuring a Quad-core Arm Cortex-A53 Application Processor Unit and a Dual-core Arm Cortex-R5 Real-Time Processor Unit [28].

Additionally this device features two key ‘hardened’ resources, i.e. integrated, non-reprogrammable cores optimised for specific tasks. For the ZU48DR chip this includes hardened RF-capable data converters, four RF-ADCs with a maximum sampling rate of 5.9 Giga Samples per Second (GSPS) and two RF-DACs capable of 9.8 GSPS sampling.

Alongside integrated data converters, the ZU48DR chip contains eight hardened user-configurable Soft Decision Forward Error Correction (SD-FEC) cores. These are integrated resources on the device which have been optimised for the encoding and decoding of Low Density Parity Check (LDPC) error detection codes as well as the decoding of turbo codes. 5G NR compatible LDPC codes are natively supported by the cores, which aids the design process and has been shown to improve performance and resource savings compared to equivalent implementations on the RFSoc PL [29].

The chosen platform for this design is the AMD UltraScale+ RFSoc4x2 [30], which features the ZU48DR RFSoc chip. Fig. 4 shows

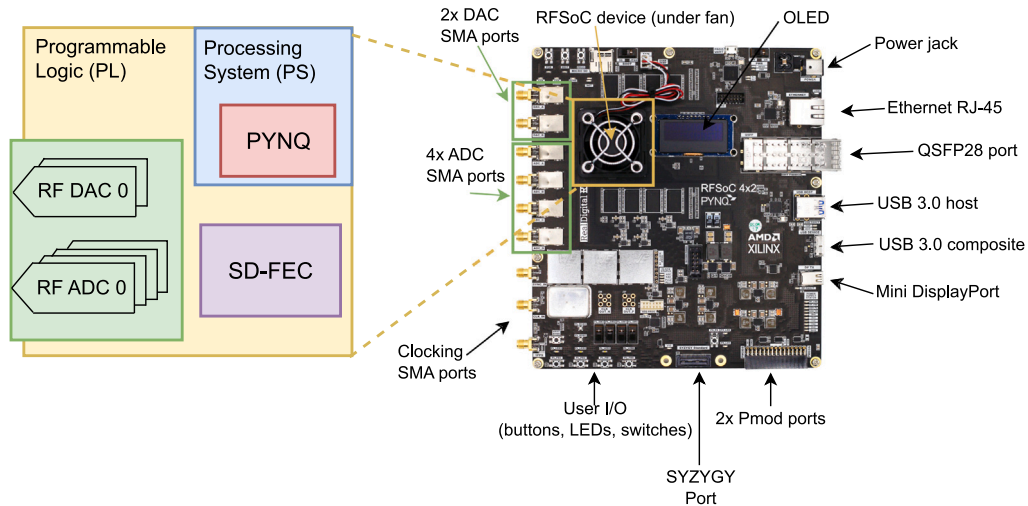


Fig. 4. Overview of the RFSoc4x2.

some of the additional functionality of the RFSoc4x2 board alongside a high level block diagram of the ZU48DR device.

1.3.2. PYNQ

The RFSoc family of devices is supported by the PYNQ framework [31], an open-source hardware/software framework from AMD. PYNQ uses the Jupyter notebook environment [32], running in the browser of a PC connected to the RFSoc, as the front end. Jupyter notebooks combine markdown and code cells, together with direct control of the PL design through *overlay* Python objects tied directly to the hardware design.

PYNQ therefore allows for ease of control on a register level or data transfer to and from the hardware design. This can be aided by the inclusion of Python ‘widgets’ from the *IPyWidgets* package [33], user input/output elements such as text boxes and dropdown menus which can be used to update a hardware design in real time. Widgets can be paired with data visualisation through graphing packages such as *plotly* [34], thus providing feedback to a user directly from the PL.

1.3.3. RFSoc PHY RU

This work uses both the PL and PS aspects of the ZU48DR chip, alongside the hardened SD-FEC cores to implement a single platform RFSoc PHY RU. The entirety of the PHY layer for the 5G NR data channel has been implemented on RFSoc hardware, with crucial LDPC encoding using the SD-FEC cores. The RFSoc PHY design is paired with a control and visualisation test harness enabled by the PYNQ framework. The result is presented as a single platform solution which can generate and process data channels for any size of transport block as initiated by a user from simple widget-based inputs. This serves to not only showcase the RFSoc as an ideal candidate for Split-6 small cell deployment, but also introduces an approach to testing radio systems using a lower functional split without the problems associated with mismatching configuration files.

1.4. Related works

Functional splits, and the application of RFSoc in this field, are well studied in both academic and industry literature. In this section we review prior work relating to each of these two topics.

1.4.1. Functional splits

One particularly noteworthy study of functional splits was performed in [35]. Here, the authors produced a thorough study of the research undergone on every functional split up to 2019, Split-1 through Split-8, as well as the feasibility of each. For Split-6, they concluded that there are strict delay requirements as time critical procedures for each RU comes from a centralised CU pool, however there are major savings in bitrate. Split-6 is shown to be particularly useful in small cell deployments where a simple DU is particularly important, which our work aims to replicate through the test harness implementation in place of a DU.

There are several pieces of literature which focus on the fronthaul benefits of Split-6 compared to other functional splits. The authors of [36] developed an Ethernet-based fronthaul test bench with a PHY/MAC split for the 4G LTE standard which achieved significant reductions in the required fronthaul bitrate compared to Split-8. The authors in [37] explore fronthaul interfaces across functional splits and note that for Split-6 MIMO systems there is improved spectral efficiency and throughput when transmitting multiple data streams over the same time–frequency resources, known as multiplexing gain, compared to other splits. The same authors in the more recent [38] propose a flexible RAN methodology to maximise spectral efficiency across fronthaul traffic. They explore a series of coordinated multi-point techniques which enable multiple base stations to coordinate, improving network performance. A Split-6 network enabled them to use a transport block selection scheme which is shown to be most effective in a small cell, single RU network.

Several recent papers have focused on feasibility studies for Split-6 deployment. In [39], the authors demonstrate the reliability of Split-6 fronthaul across a packet switch network separating the MAC and PHY layers. They implement three instances of OAI in total, a CU/DU stack with the MAC layer, a PHY layer OAI transmitter and a OAI UE, the latter two connected to USRP devices. In [40], the authors demonstrate a system which can switch between two functional split architectures at run-time, Split-6 and Split-2 (where the DU handles RLC, MAC, and PHY functionality). However, their study considers only of a split in tasks between a virtualised DU and CU, with no RU implementation. The authors of [41] present a comprehensive study of O-RAN architectures with a focus on network slicing, splitting one physical network into several isolated virtual networks. They note that Split-6 is particularly relevant for small cell deployments where per-slice level MAC control can be crucial. In [42] the authors present a digital twin model for Millimetre Wave (mmWave) and sub-THz frequencies, which concludes that Split-7.2 and Split-6 are promising

candidates for point-to-point fronthaul at extremely high frequencies. As such our work can build upon, and further prove, the feasibility of a Split-6 RU.

In addition to research papers, Split-6 can be seen in real-world testing through industry collaborations with the SCF. Examples include streetCell [43], a SCF and Dense Air co-authored project aimed at the deployment of scalable clusters of small cell networks to provide private 5G or cover gaps in MNO coverage. The SCF have also partnered with the O-RAN Alliance to develop the nFAPI specification and related Split-6 specifications for 5G and 6G [44]. This collaboration represents a major step towards the acceptance of network splits beyond Split-8 and Split-7.2 as viable deployment options.

1.4.2. RFSoc and PYNQ

RFSoc devices have proven to be popular for the realisation of Split-7.2 RUs. Examples from industry include Faststream Technologies who demonstrated an RU system using the AMD RFSoc DFE platform [45] at Mobile World Congress 2023 [46]. Research such as [47] has demonstrated how multiple 5G NR bands can be targeted using the reconfigurable nature of the RFSoc. Similarly, [48] demonstrates how a Split-8 RFSoc RU can be used to target mmWave frequencies (between 20 GHz and 200 GHz). This work demonstrated a custom RU front end to produce signals at 29 GHz using an RFSoc4x2 board. As such, the ZU48DR chip featured on the RFSoc4x2 appears an ideal candidate for further functional split implementation.

RFSoc devices have also been popular in academic literature for the implementation of disaggregated RUs. The authors of [49] have implemented a Split-7.1 DU, a variation of the popular Split-7.2, alongside an appropriate RU as a single platform. They used a vector signal generator to simulate a UE, and implemented the uplink data channel of the DU using a ZCU102 MPSoc board. The RU is implemented using an AD9371 integrated transceiver. This Split-7.1 implementation therefore is comparable to the proposed RFSoc PHY RU, though targeting Split-7.1. In [50] a 5G NR Physical Downlink Shared Channel (PDSCH), the data carrying channel of the PHY layer, transceiver is presented targeting mmWave frequency bands using a ZCU111 RFSoc. In this design, the authors have implemented QAM modulation, subcarrier mapping, OFDM modulation and interpolation for the transmitter; and demodulation, timing synchronisation and equalisation on the receiver. Their work also includes use of the PYNQ framework to control the transmitted signal and to display captured QAM constellation data. Of particular note is [51], where the authors demonstrate HELIX, a high-speed real-time 6G experimentation platform. This RFSoc-based design implements multiple functional splits alongside a UDP based fronthaul interface. This includes a Split-6 PHY implementation, serving as a key comparison point to the work presented in this paper.

The integrated SD-FEC has proven to be a powerful resource for implementing 5G NR LDPC and as such is well suited to the presented implementation method. The authors in [52] have utilised the SD-FEC to implement a 5G NR compliant LDPC decoder, using four of the SD-FEC cores in parallel, to aid in the development of drone-mounted modems. The authors of [53] present a 'DSP Engine' which implements custom FFT algorithms alongside the hardened RF Data Converters and SD-FEC of an AMD UltraScale+ ZCU111 evaluation platform RFSoc.

PYNQ has proven to be a popular method of presenting complex 5G systems. Several existing 5G NR works have presented visualisation and control elements through the PYNQ framework which this work draws upon. These works include [54] which implement a massive MIMO simulation for mmWave, with a focus on the PYNQ framework as a visualisation tool. Similarly [55] leverages the PYNQ framework through a 5G NR network function virtualisation system.

The pre-existing literature lends confidence to the use of RFSoc devices alongside the PYNQ framework for the design and presentation of disaggregated RUs as well as demonstrating the benefits of a Split-6 network. The work presented in this paper combines many of these elements together, demonstrating the PHY layer on hardware alongside a PYNQ framework test harness.

1.5. Key contributions

Split-6 small cell networks offer benefits for network deployments compared to Split-7.2 and Split-8, namely lower bandwidths across the fronthaul link and vendor interoperability. However, this Split-6 approach is arguably under-utilised in industry and under-studied in the research community. Therefore, this work has set out a unique Split-6 radio platform to provide a low-complexity approach to small cell network design and demonstrate the feasibility of an RFSoc-based Split-6 'smart' RU.

The contributions of this paper are as follows:

- *Implementation of PDSCH PHY layer using RFSoc:* A PDSCH PHY layer design has been implemented on the PL of the target chip. To the best of the author's knowledge this is the first published RFSoc implementation to perform all processing stages of PDSCH starting from unprocessed, raw data. Complex LDPC encoding has been efficiently implemented using the hardened SD-FEC resources of the target ZU48DR chip.
- *Single-chip Split-6 test harness:* Verification of the PDSCH system without the need for additional CU/DU software or a core network. A custom control system has been implemented in PL to perform resource allocation following 3GPP standards for any dimensions of input data. In addition the PYNQ framework has been utilised to design a user interface of Python widgets for low complexity resource allocation and signal generation. The user interface has been paired with constellation and resource grid plots alongside Integrated Logic Analysers (ILAs) within the design to demonstrate the success of the PDSCH implementation.
- *RFSoc Split-6 Feasibility:* The target RFSoc chip has been demonstrated as a feasible Split-6 RU platform through low resource utilisation, power consumption and a clock rate suitable for 100 MHz bandwidth transmission. Additionally, two case studies are presented to show successful PDSCH data processing and resource allocation using the test harness.

Previous studies outlined in Section 1.4, have typically focused on partial PHY implementations, software-based evaluations such as OAI, or Split-6 feasibility studies. In contrast, this paper uniquely presents a complete PHY-layer processing chain implemented on RFSoc hardware, integrated alongside a PYNQ-based control and visualisation test framework. This work therefore presents a demonstration of RFSoc as a practical Split-6 RU platform using only a single chip without external components.

The remainder of this paper explores the RFSoc PDSCH PHY layer design and test harness in detail. Section 2 introduces key background elements of the PHY layer, Section 3 discusses the IP cores in detail and Section 4 shows the hardware implementation and resource utilisation of the design. Section 5 introduces the PYNQ framework hardware/software co-design elements, and presents two case studies of the hardware design operating in conjunction with the control and visualisation systems. Finally, Section 6 summarises the results and contributions, before Section 7 introduces some of the paths forward from this work.

2. Background

The work presented in this paper focuses on the implementation of the 5G NR PHY layer in hardware with a corresponding test harness. The presented design features elements of the 3GPP 5G NR standard, namely ETSI 138_212 [56], which defines multiplexing and channel coding standards, and ETSI 138_214 [57], which defines PHY layer procedures including data transfer. The relevant methods, as standardised within these two documents, have been implemented in the RFSoc design.

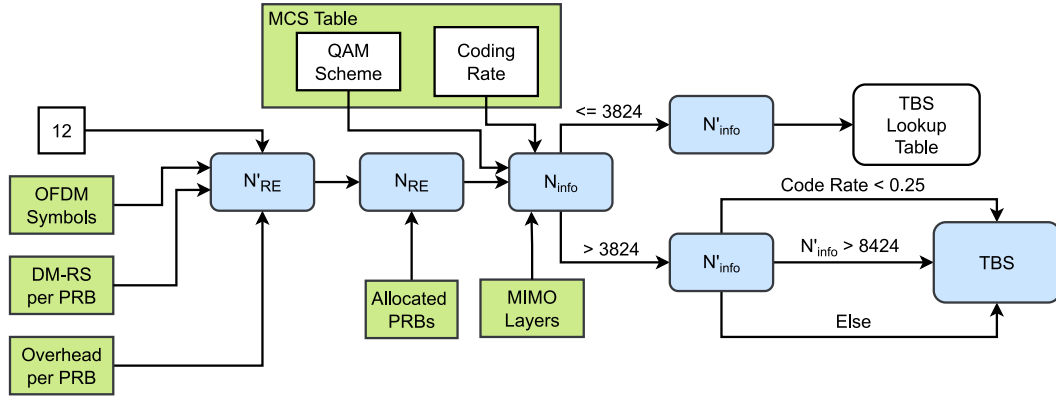


Fig. 5. Overview of transport block calculation sub-steps, indicating user inputs (green) and calculations (blue).

2.1. Transport block

In a 5G NR network, the amount of data passed on each slot in a radio frame is defined as a Transport Block (TB) [57]. Unlike a traditional S-RU, this work introduces a hardware control system and signal generation on the RFSoc PL. As such, the calculation of the Transport Block Size (TBS) is relevant to the radio design instead of a function of the DU. This section will introduce the calculations and algorithms that are required to select a TBS. Fig. 5 provides an overview of how user inputs relate to the various calculation stages. The TBS for any given slot within a radio frame can be calculated from a set number of control parameters:

- The total MIMO layers, v
- The number of allocated Physical Resource Blocks (PRBs), n_{PRB}
- The number of OFDM symbols scheduled for data within a slot, N_{symb}^{sh}
- The number of Resource Elements (REs) allocated for Demodulation Reference Signal (DM-RS), N_{DMRS}^{PRB}
- The allocated overhead, N_{oh}^{PRB}
- The Modulation and Coding Scheme (MCS) table index, giving the modulation order Q_m and the coding rate R

The number of REs within a single PRB allocated to the data channel are found as N'_{info} . To obtain this value, the total number of REs across all PRBs is first calculated, shown in Eq. (1a) as N'_{RE} . The maximum number of data bits per PRB is 156, which corresponds to thirteen symbols, with twelve subcarriers (or one PRB) in each symbol. This ensures that at least one symbol in each slot is reserved for other channels. Hence, N_{RE} can be found as the minimum of 156 and N'_{RE} , as in Eq. (1b).

$$N'_{RE} = N_{SC}^{RB} \cdot N_{symb}^{sh} - N_{DMRS}^{PRB} - N_{oh}^{PRB} \quad (1a)$$

$$N_{RE} = \min(156, N'_{RE}) \cdot n_{PRB} \quad (1b)$$

The total number of information bits can be calculated as Eq. (2), based on the value of N_{RE} .

$$N_{info} = N_{RE} \cdot R \cdot Q_m \cdot v \quad (2)$$

Depending on the value of N_{info} , one of two methods are applied to find the quantised number of information bits N'_{info} . If $N_{info} \leq 3824$, (3a) is used, otherwise (3b) is applied.

$$N'_{info} = \max \left(24, 2^n \left\lceil \frac{N_{info}}{2^n} \right\rceil \right), \quad \text{where } n = \max(3, \lfloor \log_2(N_{info}) \rfloor - 6) \quad (3a)$$

$$N'_{info} = \max \left(3840, 2^n \times \text{round} \left(\frac{N_{info} - 24}{2^n} \right) \right),$$

$$\text{where } n = \lfloor \log_2(N_{info} - 24) \rfloor - 5 \quad (3b)$$

Following this, if $N_{info} \leq 3824$ then Table 5.1.3.2-1 of [57] is used to find the closest TBS that is greater than N'_{info} . Otherwise, the value of TBS is calculated from one of Eqs. (4a), (4b) or (4c) based on the code rate and N'_{info} .

$$\text{if } R \leq \frac{1}{4}, \quad TBS = 8 \cdot C \left\lceil \frac{N'_{info} + 24}{8 \cdot C} \right\rceil - 24, \quad \text{where } C = \left\lceil \frac{N'_{info} + 24}{3816} \right\rceil \quad (4a)$$

$$\text{else if } N'_{info} > 8424, \quad TBS = 8 \cdot C \left\lceil \frac{N'_{info} + 24}{8 \cdot C} \right\rceil - 24, \quad \text{where } C = \left\lceil \frac{N'_{info} + 24}{8424} \right\rceil \quad (4b)$$

$$\text{else } TBS = 8 \left\lceil \frac{N'_{info} + 24}{8} \right\rceil - 24 \quad (4c)$$

2.2. PHY layer

The 5G NR PHY layer comprises of four main channels: the PDSCH, Physical Downlink Control Channel (PDCCH), Physical Random Access Channel (PRACH) and Physical Broadcast Channel (PBCH). These are combined together via a multiplexer, before OFDMA symbols are generated. This work focuses on the implementation of PDSCH, the data channel of the PHY layer. PDSCH is comprised of several sub-steps as can be seen in purple in Fig. 6, focused primarily on bit error mitigation to ensure reliable transmission over a wireless channel. These stages are standardised by 3GPP in ETSI 138.212 [56]. The following subsections will introduce some of the key considerations required for successful data processing, which have been implemented on the RFSoc4x2.

The main purpose of processing the raw data bits through PDSCH is to add FEC for reliable transmission. Within 3GPP standardisation for 5G NR, the common approach to this is LDPC encoding. The transmitter PHY layer can therefore be discussed in terms of LDPC pre-processing and post-processing.

The letter annotations in the following subsection titles refer to blocks shown in Fig. 6. Note that we begin with block (C) Error Correction to simplify the explanations.

2.2.1. Error correction (C)

LDPC operates on a principle of predefined codewords, c , which satisfy a sparse parity matrix, H , i.e. the matrix contains considerably more 0s than 1s. Each row in this matrix represents an XOR based parity

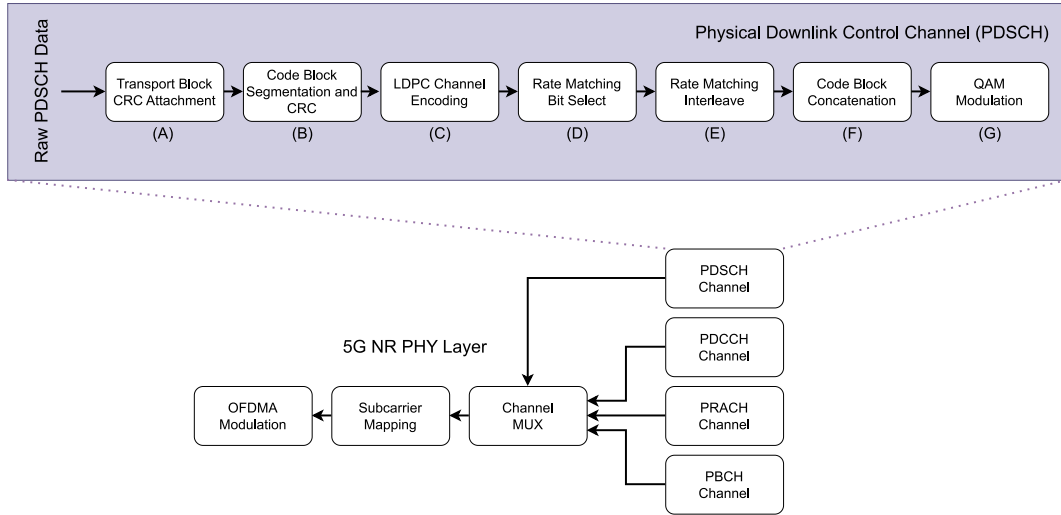


Fig. 6. Overview of PDSCH PHY layer transmitter data processing sub-steps.

equation, while each column represents one bit in the codeword. From this, based on input data, parity bits are found which satisfy $H \cdot c^T = 0$. This approach is well suited to hardware deployment, using quasi-cyclic codewords which enables encoding using linear feedback shift registers.

The LDPC matrix, which can be extremely large and difficult to implement, can instead be represented through a Base Graph, a compact matrix where each of the entries can be further expanded. As such, considerably fewer resources are required to store the H matrix. 5G NR supports two Base Graph options, each with a predefined size; Base Graph 1 (BG1) is a 46×48 matrix designed for large TBs, whilst Base Graph 2 (BG2) is a 42×52 matrix for smaller TBs.

These Base Graphs can be expanded through a factor Z_c , referred to as the lifting size variable. Each entry in the Base Graph can be 'lifted' into a $Z_c \times Z_c$ matrix to reform the parity-check H matrix. Table 5.3.2-1 of [56] defines discrete Z_c values which are indexed by TBS. As such, a 5G NR LDPC encoder can be configured with pre-defined codewords and input with a Base Graph and Z_c value for any TBS.

2.2.2. Cyclic redundancy check (A) and segmentation (B)

To perform successful LDPC encoding, pre-processing must occur on the PDSCH data. A CRC is initially attached to the TB to aid in the detection of errors with the received data block. The maximum amount of data which can be processed by the LDPC in a single process, with BG2, is 3,840 and therefore if the number of data bits plus the CRC is less than 3,840 a single LDPC process can be used. Therefore, the length of the CRC is dependant on the TBS value, if greater than 3,824 a 24-bit CRC from the cyclic generator polynomial $G_{CRC24A}(D)$ is appended. Otherwise, a 16-bit CRC from the generator polynomial $G_{CRC16}(D)$ is attached.

If the number of data bits is greater than 3,824 it is desirable to segment the TB into equally sized code blocks. The maximum size of these code blocks is determined by the Base Graph used for LDPC encoding, a representation of the LDPC parity matrix. The decision over which Base Graph is used depends on a threshold of both TBS and Code Rate.

$$\begin{aligned} TBS \leq 292 & \rightarrow \text{LDPC base graph 2} \\ TBS \leq 3824 \text{ and } R \leq 0.67 & \rightarrow \text{LDPC base graph 2} \\ R \leq 0.25 & \rightarrow \text{LDPC base graph 2} \\ \text{else} & \rightarrow \text{LDPC base graph 1} \end{aligned}$$

For BG1 the maximum size of a code block is 8,448, whilst for BG2 this maximum size is 3,840. If the TBS value is greater than the maximum code block size K_{cb} , Eq. (5) can be used to find the number of segmented code blocks C and the number of bits per code block K'

where B is the number of bits following the initial appending of the CRC, L is the length of the CRC, and B' is number of bits following code block CRC appending, i.e. an additional 24-bits per code block.

$$B = TBS + L \quad (5a)$$

$$C = \left\lceil \frac{B}{K_{cb} - 24} \right\rceil \quad (5b)$$

$$B' = B + 24C \quad (5c)$$

$$K' = \frac{B'}{C} \quad (5d)$$

LDPC encoding within 5G NR expects specific input sizes depending on the Base Graph and consequently, the code blocks are padded with NULL values. The LDPC block size is based on the lifting size Z_c , chosen from a set of possible values in Table 5.3.2-1 of [56] such that $Z_c \geq \frac{K'}{K_b}$ where $K_b = 22$ for BG1 or $K_b = 10$ for BG2.

Therefore each LDPC code block is K' bits of the TB followed by a 24-bit CRC obtained from the cyclic generator polynomial $G_{CRC24B}(D)$. Each code block is then padded with NULL values from K' to $K = 22Z_c$ for BG1 or $K = 10Z_c$ for BG2. The concept of CRC appending, code block segmentation and padding can be further seen in Fig. 7. Here an example is presented, showing the steps necessary to obtain padded code blocks from a given TBS of 19,968 bits.

2.2.3. Rate matching (D, E)

The output size of the LDPC sequence for each code block is considerably greater than the allocated transmission resources. A process is required through which bits are extracted from the LDPC code blocks to fill the initial allocated resource elements with QAM modulated symbols. Initially, all encoded data and parity bits from the LDPC process are collected into a circular buffer. For each code block in turn, a number of bits are extracted from the buffer, determined by the sequence length E_r ,

$$E_r = N_L \cdot Q_m \cdot \begin{cases} \left\lfloor \frac{G}{N_L \cdot Q_m \cdot C'} \right\rfloor, & \text{if } j \leq C' - \text{mod} \left(\frac{G}{N_L \cdot Q_m}, C' \right) - 1 \\ \left\lfloor \frac{G}{N_L \cdot Q_m \cdot C'} \right\rfloor, & \text{otherwise} \end{cases} \quad (6)$$

where j represents the current code block, N_L is the number of transmission layers, Q_m is the modulation order, G is the total number of data bits ($RE \cdot Q_m$) and C' is the number of code blocks. Note that the

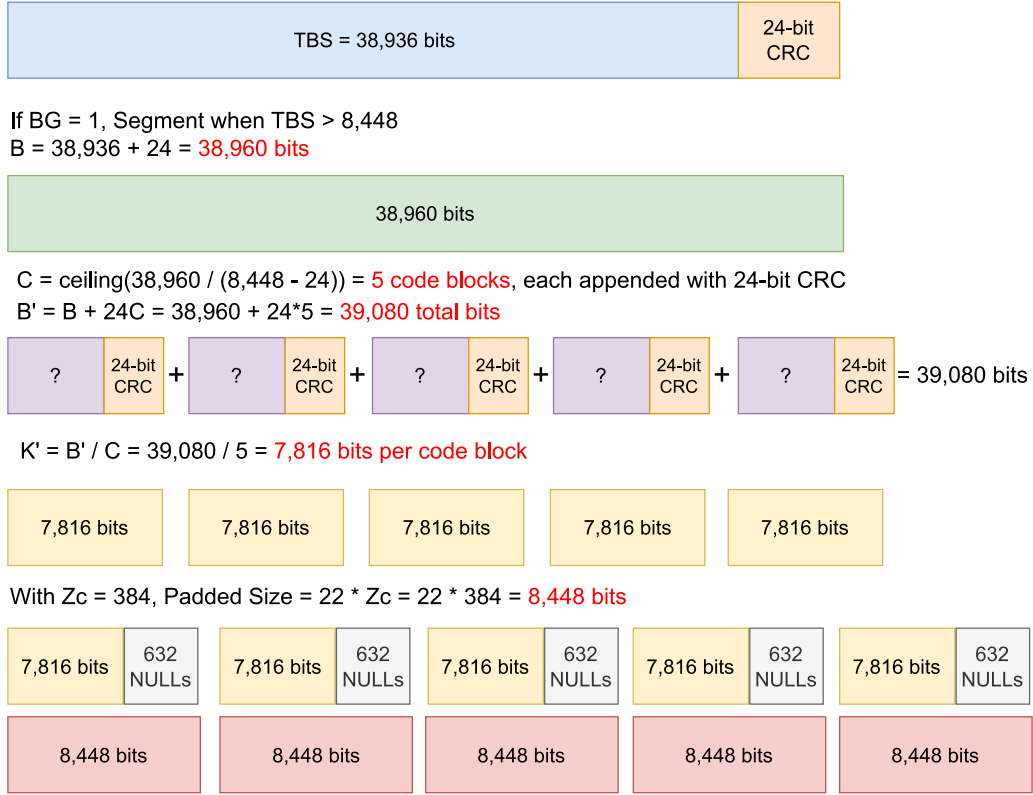


Fig. 7. Code block segmentation example for a given TB of 38,936 bits.

only difference between the pair of E_r equations is the use of ceiling and floor operators, which ensure that the combined total of bits N_b across all code blocks matches $N_b = N_L \cdot Q_m \cdot N_{RE}$. The rate matching output sequence, e_k can then be determined from Algorithm 1.

Algorithm 1 LDPC Bit Extraction

```

1: while  $k < E_r$ 
2:   if  $d_{(k_0+j) \bmod N} \neq \langle \text{NULL} \rangle$ 
3:      $e_k = d_{(k_0+j) \bmod N}$ 
4:   end if
5: end while

```

In algorithm 1, $N = 66Z_c$ or $N = 50Z_c$ for BG1 and BG2 respectively, k and j are initialised as zero and increment by one each iteration, and d is the number of LDPC output data bits. The value of k_0 determines the starting position within the LDPC output for different redundancy versions based on an rv_{id} input as determined by Table 5.4.2.1-2 of [56]. This algorithm therefore ensures that an appropriate E_r number of bits from a specified starting location within the LDPC code block are extracted to fill all ER of the output resource grid with QAM data.

Following bit selection, the bits are interleaved according to Algorithm 2 to improve robustness, where E_r is the output sequence length, Q_m is the modulation order, e is the input data from Algorithm 1 and f represents the interleaved output data.

2.2.4. Concatenation (F) and modulation (G)

Code blocks are concatenated together into a single data stream before modulation. This is typically a sequential concatenation in the same order as the original transport block. The 5G NR standard supports four modulation schemes in normal operation; QPSK, 16-QAM,

Algorithm 2 Bit Interleaving

```

1: for  $j = 0$  to  $\frac{E_r}{Q_m} - 1$ 
2:   for  $i = 0$  to  $Q_m - 1$ 
3:      $f_{i+j \cdot Q_m} = e_{i \cdot \frac{E_r}{Q_m} + j}$ 
4:   end for
5: end for

```

64-QAM and 256-QAM. The selected scheme is traditionally updated per radio frame as a new MCS table index is selected. The use of QAM modulation ensures that each resource element will contain multiple bits of data, greatly increasing the data rate of transmission.

The process of rate matching, concatenation and QAM modulation is illustrated with an example in Fig. 8, where RE = 15,162 bits and 16-QAM modulation is desired.

3. System modelling

The proposed RU design features a number of custom hardware IP cores, along with the hardened SD-FEC resources, and a PYNQ-based transport block control and visualisation system. This section will discuss the function of the custom IP core models, and how they relate to the equations presented in Section 2. A block diagram is presented for each. The custom IP cores are split between the unique control system which replicates some of the functionality of the MAC layer, and the PHY layer implementation.

3.1. Design methodology and interfacing

The RFSoc PHY RU has been realised through Vivado IP Integrator [58], an Integrated Development Environment (IDE) from AMD

LDPC Encoding outputs are much larger than the allocated REs.
In this example, $RE = 15,162$ bits, $Q_m = 4$ (16-QAM)

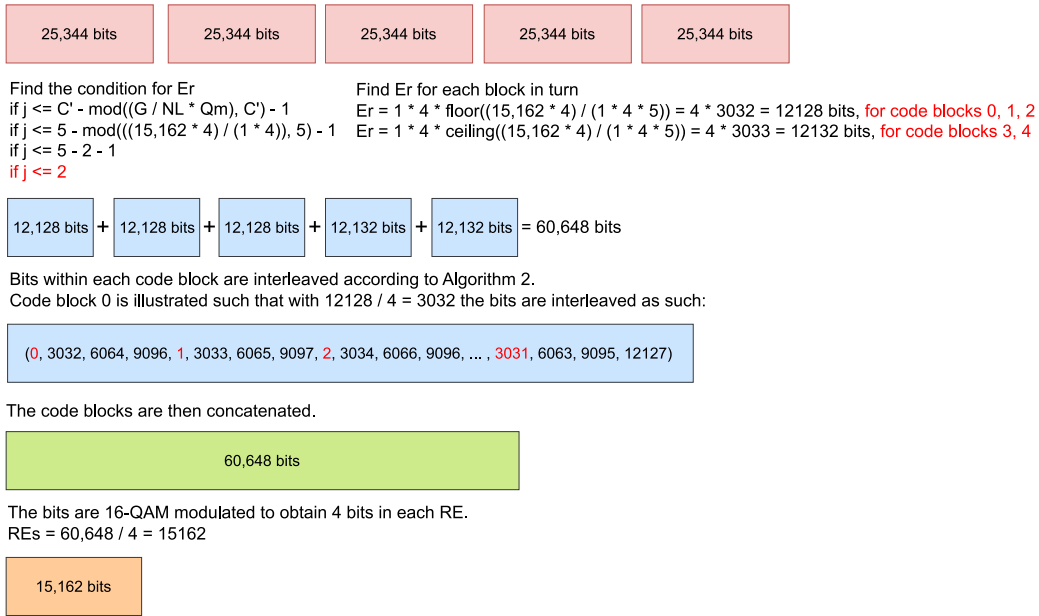


Fig. 8. Example of rate matching, concatenation and QAM modulation following LDPC Encoding.

which simplifies complex hardware designs by connecting and configuring blocks of reusable functionality as IP cores. Additionally, chip-specific hardened resources such as the RF Data Converters and SD-FEC can be accessed as if they were IP Cores, aiding in the design process.

Custom functionality was designed for the PL using the HDL Coder [59] block-based design tool from MathWorks. This design method is useful for communications systems, as it offers features for precise control of data movement, coordination logic, and arithmetic features, as well as a rich set of modelling and simulation functionality.

The movement of data through an IP Integrator design is standardised as AXI4 [60]. This work features two of the signal types contained within the AXI4 standard; AXI4-Lite and AXI4-Stream. The former is a memory-mapped, single transaction interface well-suited to register level configuration of PS-to-PL connections. AXI-Lite is mainly used in the presented design to automate initialisation of the PHY layer IP Cores based on the calculated TBS and related parameters. AXI4-Stream enables the direct flow of data between IP cores in PL, and is used to pass PDSCH data between processing stages.

The Vivado IP Integrator design has been realised through the creation of six custom IP cores; a *Resource Allocation Controller*, *Signal Generator*, *CRC Append and Code Block Segmentation*, *SD-FEC Controller*, *Bit Selection*, as well as *Bit Interleave and QAM Modulation*. A high level overview of the HDL Coder design can be seen in Fig. 9. The colours used in Figs. 10–15 correspond to the blocks in Fig. 9.

Note that the SD-FEC is hardened functionality of the RFSoc and therefore does not require to be designed in HDL Coder. A copy of the signal generation IP core, with customised input logic, has been added to generate the same number of bits per code block as the SD-FEC would for the same control inputs.

3.2. Resource allocation controller

To drive a single platform S-RU test unit, a control model was created. This model is responsible for the calculation of resource element allocation and transport block size, based on user input. These calculated values then automatically select an LDPC base graph, the number of code blocks, bits per code block and the LDPC lifting size. The model enables accurate simulation of the PHY layer models as well as enabling a test harness for the hardware implementation. The

outputs from this model are written to AXI4-Lite registers which can be read from other IP cores as required. In this way, with a few simple inputs, the PHY channel can be fully automated. As such, the controller IP core replicates functionality of an S-DU, producing the required size of transport block and passing relevant control channel information to the stages of PHY layer processing.

On initialisation, user input from PYNQ provides high level control channel information about radio frame resource allocation. These inputs include the total number of OFDMA symbols allocated to the transmission of data, how many resource blocks each symbol contains and which modulation scheme and coding rate will be used (selected from one of three tables). As in Fig. 10, the system controller IP Core is passed these parameters via an AXI4-Lite protocol interface from PYNQ and outputs parameters which configure the remainder of the system. The main purpose of this IP core is to obtain the TBS, a measure of the number of information bits to be transmitted. These calculations are performed in hardware and are based on the formulas specified in Eq. (4).

3.3. PHY layer

The implementation of the PDSCH PHY layer comprises several IP cores, which are presented through the following subsections. Five custom IP cores have been created for PHY layer processing which can be categorised as pre-LDPC processing (signal generation, CRC addition and code block segmentation, SD-FEC control logic) and post-LDPC processing (bit selection, interleaving and modulation).

3.3.1. Signal generation

In a traditional Split-6 radio system, TBs of data would be received by the RU over Ethernet, through eCPRI. In this work, the RFSoc acts as a single contained system and therefore a signal generator based on the TBS is required as shown in Fig. 11. This model takes four AXI4-Lite inputs, *aximm_enable*, *aximm_bits*, *aximm_frames*, and *aximm_reset* and outputs an AXI4-Stream of samples. The *aximm_enable* signal triggers a rising edge detector that provides a trigger to a free-running counter, which is reset when it reaches the value of the transport block size, *aximm_bits*. This counter increases in increments of eight, producing an *m_axis_tvalid* control signal eight times shorter than the transport block

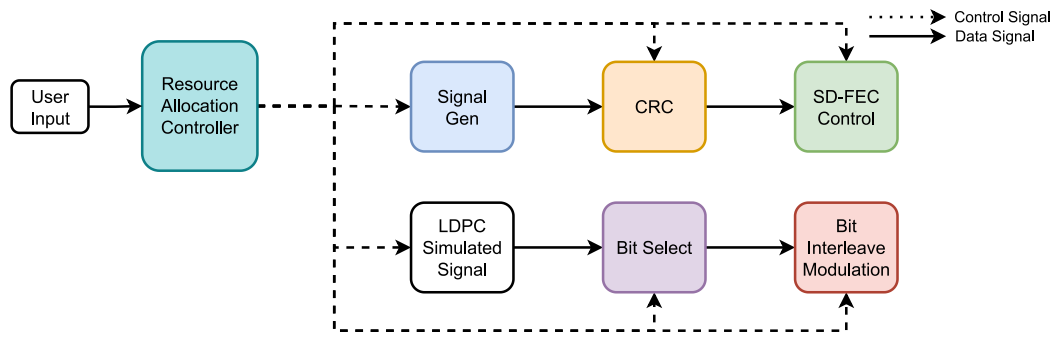


Fig. 9. High level overview of HDL coder models.

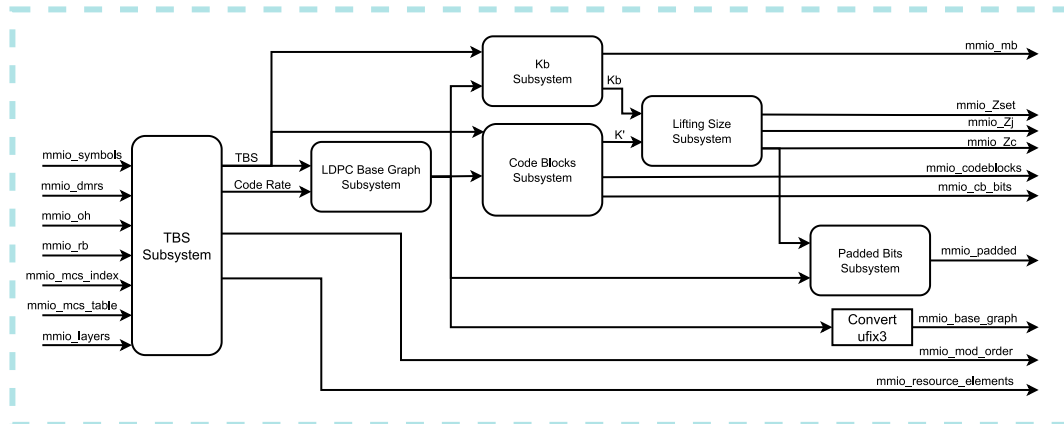


Fig. 10. Block diagram of the S-RU controller HDL coder model.

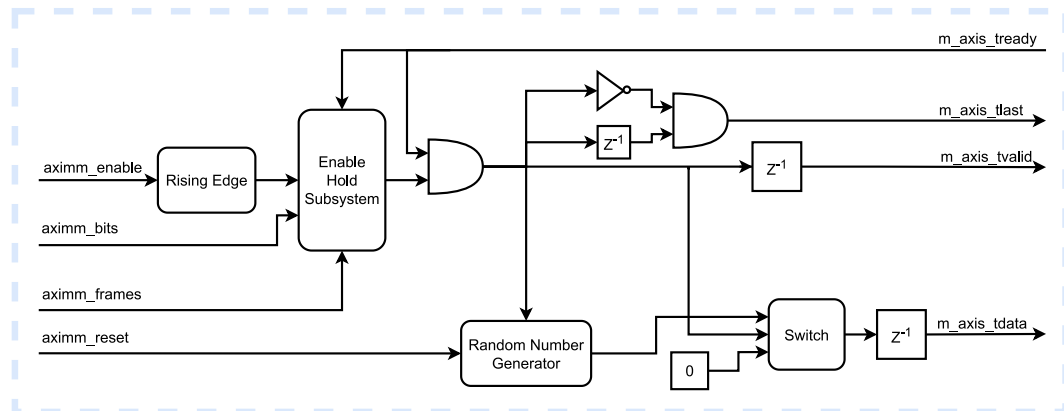


Fig. 11. Block diagram of signal generation HDL coder model.

size. This drives a uint8 random number generator with the polynomial $z^{13} + z^{12} + z^{11} + z^8 + 1$. The value of *aximm_slots* determines how many times this control signal is repeated.

3.3.2. CRC and segmentation

Following the generation of data, an IP core appends a CRC to the whole of the TB and segments the data into code blocks as required. If the TB is segmented an additional 24-bit CRC is appended to each code block, before the data is padded prior to performing LDPC encoding.

A high level view of this system can be seen in [Fig. 12](#). CRC attachments are performed using the *NR CRC Encoder* HDL Coder block. A CRC of 16 or 24 bits is initially appended to the TB dependant on the TBS input, *aximm_tbs*. To segment the data, it is buffered into a FIFO and read out in code blocks of a size determined by the *Resource*

Allocation Controller. If the number of code blocks is greater than one, an additional *NR CRC Encoder* is used to attach a 24-bit CRC to each code block. The data bits are then buffered with an additional FIFO, where they are read out appended with NULL values to match the size expected by the hardened SD-FEC core. To achieve this, combinatorial control logic and a counter are used to produce an *m_axis_tvalid* signal of the padded length.

3.3.3. SD-FEC controller

For each block of data supplied to the SD-FEC IP core, a control word requires to be supplied on its CTRL interface, one sample before the data block begins. For the 5G NR LDPC code, this control word consists of 40 bits, and comprises the fields shown in Table 1. If a control word is not supplied the core will not process data and will

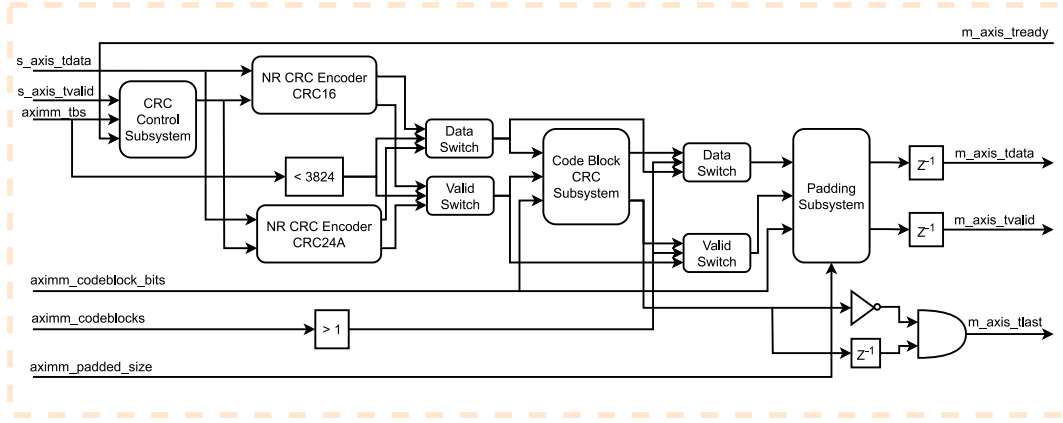


Fig. 12. Block Diagram of CRC and Code Block Segmentation HDL Coder Model.

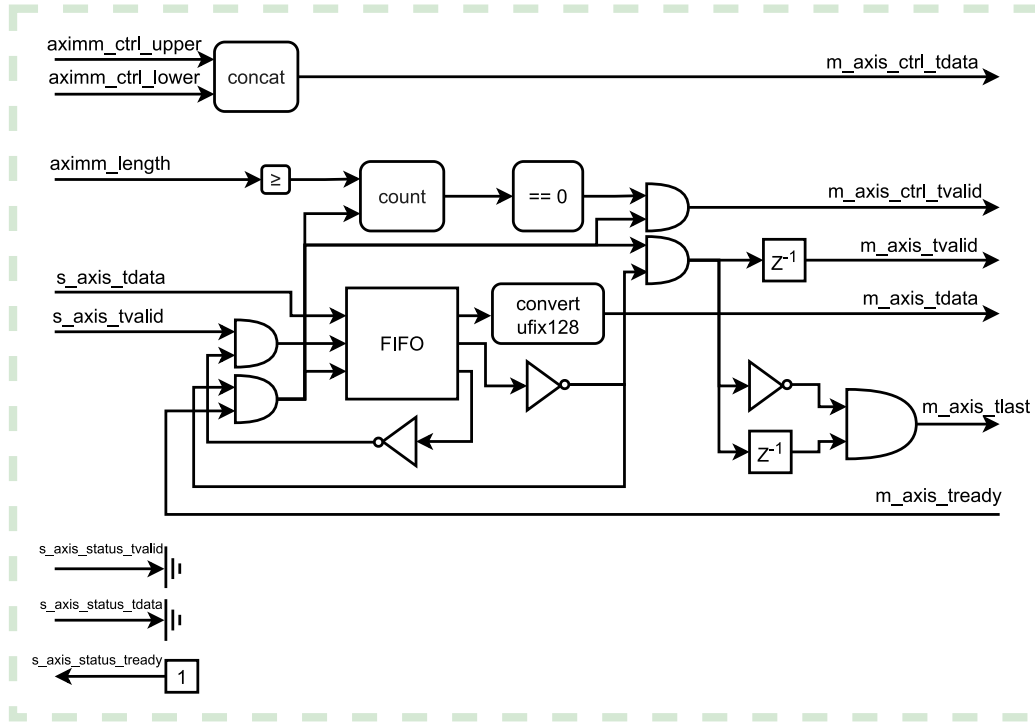


Fig. 13. Block Diagram of the Control System HDL Coder Model for the hardened SD-FEC.

Table 1
SD-FEC Control Words for 5G NR.

Field (bits)	Type
Max Schedule (39:38)	ufix2
Parity Bits (37:32)	ufix6
External Block Identifier (31:24)	uint8
Reserved (23:9)	ufix15
Base Graph (8:6)	ufix3
Base Graph Cyclic Set (5:3)	ufix3
Lifting Factor (2:0)	ufix3

prevent future data from being input. Similarly, a status sample must be read from the SD-FEC core or it will not process any future code blocks.

The data channel, s_axis_tdata , is buffered using a FIFO, with a valid signal read from the m_axis_tready value of the SD-FEC to prevent stalling. The valid signal additionally triggers the enable of a free running counter which counts up to the length of a code block,

determined by the AXI4-Lite $aximm_length$ input, before being reset. A single sample control valid is produced upon a reset of this counter. The control data is read from two AXI4-Lite registers, $aximm_ctrl_upper$ and $aximm_ctrl_lower$, before being concatenated due to the uint32 size limit of AXI4-Lite. A single valid AXI4-Stream control sample is generated at the start of each block of data.

3.3.4. Bit select

A process is required to extract a variable number of bits from the code blocks which have been extended with parity bits following the SD-FEC core. This IP core can be seen in Fig. 14, showing a FIFO to buffer data and control logic to produce a valid signal of a length specified by Algorithm 1. An E_r value is calculated from three AXI4-Lite interfaces; $mmio_resource_elements$, $mmio_mod_order$ and $mmio_codeblocks$. Similarly, the redundancy value rv_{id} value is input from AXI4-Lite, and is used as the control input to multiplexers to obtain a starting position k_0 value. Using a free running counter, a valid signal can be generated for the length of one LDPC code block, either

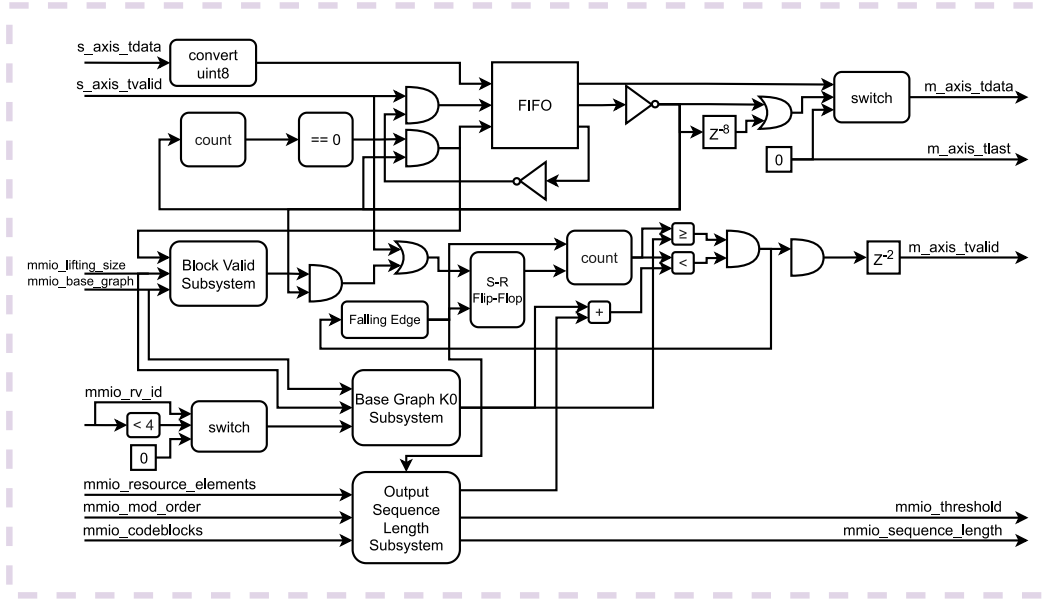


Fig. 14. Block diagram for the bit select HDL coder model.

$N = 66Z_c$ or $N = 50Z_c$ for BG1 or BG2, respectively. The rising edge of this valid signal triggers an S-R flip-flop. This enables a free-running counter, which produces a further valid signal for values between k_0 and $k_0 + E_r$. The falling edge of this valid resets the S-R flip-flop. In parallel with this valid signal generation process, the data is buffered in a FIFO after conversion from uint128 to uint8, following the output of the SD-FEC *tdata* AXI4-Stream. This value is read from the FIFO as a uint8 on each eighth valid produced by the free-running counter, to simplify the interleaving process which requires individual bits.

3.3.5. Interleave and modulation

An additional module is created to contain the remaining element of post-LDPC rate matching, i.e. bit interleaving, as well as an implementation of QAM modulation. As can be seen in Fig. 15, at a high level this IP core reads data from a Dual-Port RAM block in an interleaved pattern, before one of four modulation schemes are automatically selected. Finally a FIFO is used to adjust the sample rate respective to the modulation scheme.

The input data *s_axi_tdata* consists of a uint8 value on each eighth sample. The interleaving process described in Algorithm 2 is performed on a per-bit instance, so a bit slice operation is applied to the data channel, allowing ufix1 samples from each valid sample to be written to a Dual-Port RAM. A control system comprising of counters, adders and multipliers generates the values of f and e from Algorithm 2, representing the write address and read address of the Dual-port RAM, respectively. The latency of reading from the Dual-Port RAM is variable, based on the sequence length and modulation order, and to accommodate this the read addresses are buffered using a FIFO.

The modulation schemes are applied in parallel subsystems before a switch determines which is valid for the current modulation order. These subsystems are similar, consisting of two stages, the first a bit concatenation of the appropriate number of samples, e.g. two for QPSK, eight for 256-QAM. This is followed by a QAM generator featuring multiport switches for I and Q channels which map individual bits to voltage levels, e.g. 1 and -1 for QPSK before concatenating the bits once more. Appropriate valid samples for the respective modulation schemes are generated using a counter and comparison logic blocks. The final stage of operation is to reduce the sample rate using a FIFO so that a stream of constant valid samples can be output from this core to the subsequent IFFT to obtain OFDM symbols.

4. Hardware results

A block diagram of the S-RU RFSoc implementation can be seen in Fig. 16. The baseband signal is clocked at 122.88 MHz, generated from the RF-DAC clock, which is representative of a 100 MHz bandwidth signal. The design meets implementation timing at this clock rate, achieving a Worst Negative Slack (WNS) of 0.595 ns. As such, this design can be flexibly used to generate data appropriate for any sub-6 GHz 5G NR band through decimation.

Power measurements have been taken from the implemented design, giving the total ZU48DR on-chip power usage as 4.33 W, consisting of 2.533 W dynamic power and 1.276 W static power. This can be compared against a power-saving, small cell targeted RU from Mavenir and Intel which consumes around 100 W [61]. Similarly, real-world 64x64 Massive MIMO 5G NR deployments can consume 1 to 2 kW of power [62,63]. Note that the presented RFSoc design does not represent a full RU, and antenna array and power amplifiers contribute heavily to these real-world deployments. The low initial usage of the RFSoc RU however gives confidence towards further development of a Split-6 transmitter.

Fig. 17 illustrates the estimated energy-per-bit (E_b) as derived from the 2.533 W dynamic power reported from Vivado, across standard 5G NR modulation schemes and common RB values. Three cases are shown: 51, 133 and 273 RBs which correspond to the maximum RBs allowed for a 20 MHz, 50 MHz and 100 MHz bandwidth respectively, with a 30 kHz SCS as assumed throughout this work. The results confirm the expected trend that E_b decreases with higher modulation order as more information bits are allocated per symbol. Similarly, larger bandwidth allocations reduce E_b as the same fixed dynamic power is distributed across a greater number of transmitted bits. Note that the values reported here reflect implementation-level E_b , and thus capture the cost of the digital baseband processing only, independent of RF transmission losses. The 20 MHz case exhibits the highest E_b across all modulation schemes, while the 100 MHz case is an order of magnitude lower. As E_b is a measure of the energy required to process a single information bit through the implemented design, increasing the transmission bandwidth and the modulation scheme both greatly increase the energy efficiency of this hardware implementation.

Table 2 shows the resource usage of this design, both with and without the test harness resources (presented in Section 5). The obtained results have been compared against [49], which implements a

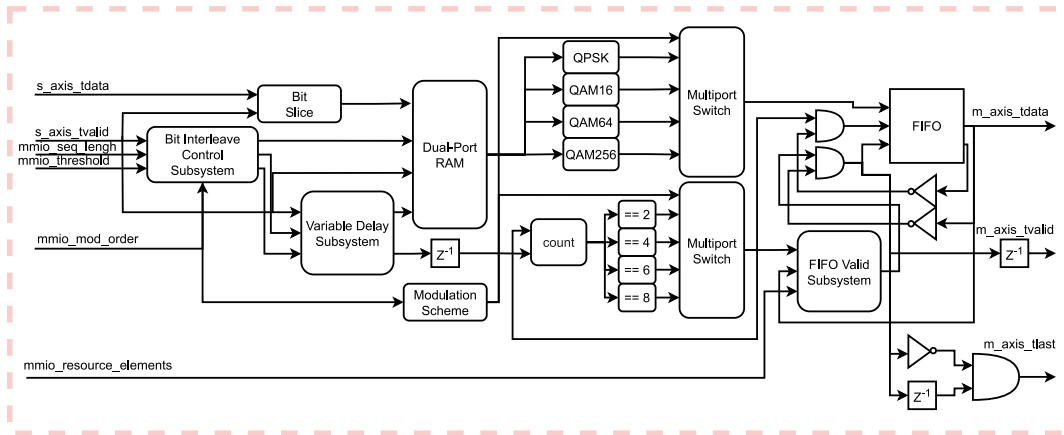


Fig. 15. Block diagram of bit interleave and modulation HDL coder model.

Table 2
RFSoc ZU48DR Chip resource utilisation.

Resource	PDSCH PHY layer		With test harness		Paper [49] DU and RU
	Used	Utilisation %	Used	Utilisation %	
LUT	10,890	2.56	25,015	5.88	51,448
LUTRAM	547	0.26	1,238	0.58	11,367
FF	12,894	1.52	29,129	3.42	70,780
BRAM	83.50	7.73	256.50	23.75	486.5
DSP	13	0.30	27	0.63	194
BUFG	4	0.57	11	1.58	62
MMCM	0	0	1	12.50	3

Split-7.1 RU and DU on a ZCU104 MPSoC coupled with an external transceiver, and is therefore similar to the work presented in this paper. It can be seen that the PDSCH PHY layer implementation (in isolation) is extremely light on resource usage, with under 10% usage of any individual resource. The HDL Coder models are designed primarily from combinatorial logic, counters, switches, etc. and, partly as a result, the Lookup Table (LUT) and Flip-Flop (FF) resources remain under 3%. The *Interleave and Modulation* IP core features a dedicated Dual-Port RAM which is mapped to Block RAM (BRAM) resources. As such BRAM usage is notably higher than other resources at 7.73%. Note the limited use of DSPs in this design, using only 0.3% of those available.

Even with the test harness IP cores included in the design, consisting of the *Resource Allocation Controller*, *Signal Generation* and Integrated Logic Analysers (ILA) for hardware, the design still uses many fewer LUT and FF resources, and only a fraction of the LUTRAM, when compared to [49]. The ILA cores can be configured to buffer large amounts of data at a time for analysis by using BRAM resources. With the design configured to store 32,000 samples at once in ILAs, the design still achieves under 25% of BRAM allocation.

The results from Table 2 give confidence that the RFSoc PHY RU and test platform can be expanded in future to support the additional PHY layer channels for a complete 5G NR radio.

5. PYNQ test harness

Key to this work is the pairing of an S-RU PHY layer implementation with user input control parameters, creating a single platform RFSoc Split-6 RU and test harness. To this end, an interactive graphic user interface (GUI) has been created through the PYNQ framework, allowing for control of the transport block generation and a resource grid visualisation using data from the PHY layer produced in hardware. The PYNQ visualisation tool displays either an IQ plot or a resource grid, using the processed PDSCH data from a Direct Memory Access (DMA) IP core, obtained upon pressing *Generate Signal* button in the GUI. To

Table 3
Test harness assumptions.

Variable	Assumed value
Number of Layers	1
Overhead	0
DM-RS Type	1
DM-RS Max Length	1
PDSCH Type	A

the right of the display, the user has input control from Python widgets. Below this is the output of the controller IP core, which updates in real time as the inputs change. Examples of the PYNQ GUI can be seen as Figs. 18–19.

User inputs are provided for the number of symbols containing data, MCS table and index, resource blocks per symbol, and the allocated slots for PDSCH data. The controller IP operates in real-time, in response to any changes to user inputs. The number of symbols is limited to the range 1–20 for demonstration purposes, however due to the minimum condition in Eq. (1b), REs are capped at thirteen symbols. The three MCS tables (64-QAM, 256-QAM and low signal quality 64-QAM) can be selected, as well as a table index, thus allowing for a manual selection of modulation scheme and code rate. The number of PRBs can be selected between 1 and 273, the maximum for a 100 MHz bandwidth FR1 signal with a 30 kHz subcarrier spacing. Finally, the number of consecutive slots to fill with PDSCH data can be selected, which will produce multiple TBs worth of bits. From these inputs, outputs are read back from the *Resource Allocation Controller* IP core in real time and presented to a user through text boxes within the GUI. These are: the number of REs allocated both for one slot and the total number of REs; the TBS as calculated by Eq. (4); and the current modulation scheme from the MCS table and index selections.

The user input/output widgets are directly connected to the hardware design via an overlay Python object. As such, when a user input field is changed, the *Resource Allocation Controller* IP core is updated in real time alongside appropriate AXI4-Lite inputs of the PHY layer IP cores. Therefore, when a user presses the *Generate Signal* button of the GUI, an amount of data bits equivalent to the TBS are produced, which is then processed by the PHY layer as described in Section 3. Data is passed to the PS via DMA, where it is used to produce an IQ plot of the modulation scheme and a resource grid. A demodulation reference signal (DM-RS) has been added to this plot, using the same assumptions as Section 7.1.2.2.4.2 of [64] to simplify the test harness resource grid plot. These assumptions are summarised in Table 3.

By assuming a single layer and no overhead, the GUI of the PYNQ test harness can be simplified whilst still giving a user control over the resource allocation. DM-RS type 1 is common for UE connections

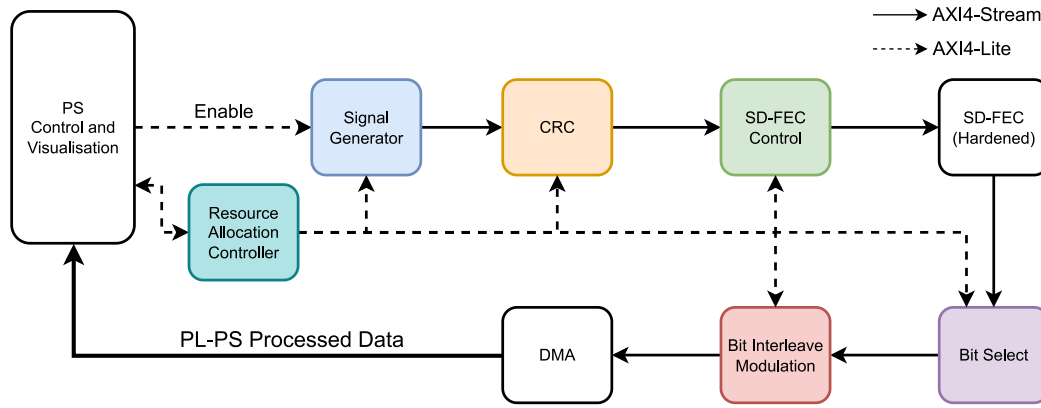


Fig. 16. Block diagram overview of Vivado IP integrator design.

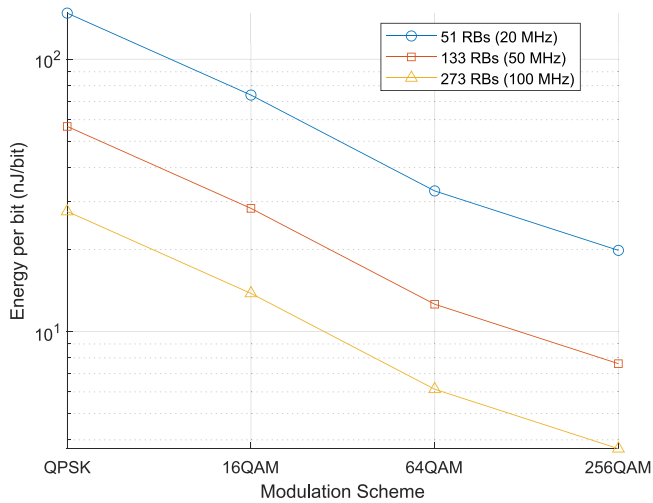


Fig. 17. E_b calculated for standard 5G modulation schemes across three common RB allocations (20, 50, and 100 MHz). Values are derived from Vivado dynamic power estimates (2.533 W) assuming 14 symbols per slot and 30 kHz SCS.

and ensures that each second subcarrier within each resource block in the allocated symbol(s) contains DM-RS. Setting the DM-RS maximum length parameter to 1 means that a single symbol is allocated to DM-RS, which results in a clear resource grid plot. Selecting PDSCH Type A means that the DM-RS symbol may occur in either symbol 3 or 4, however in this implementation it has been further assumed to always occur in symbol 3.

5.1. Implementation case studies

Two examples are presented to further explore the combination of control system and PDSCH on a single embedded platform. User input parameters and recorded *Resource Allocation Controller* IP core outputs are presented in Table 4 for both cases. The output number of bits at each stage of the PHY layer are additionally presented in Table 5 for both cases.

5.1.1. Case 1: 51 Allocated PRBs, QPSK modulation

The inputs for Case 1, which can be seen alongside measured outputs in the Case 1 column of Table 4, represent a 20 MHz cell with 30 kHz subcarrier spacing, corresponding to 51 PRBs. Thirteen symbols are allocated, the maximum that can be used for data in a single slot. We consider an illustrative example using the 64-QAM MCS table and an index of 6. In a real system, this would represent an especially poor

Table 4

Case study parameters.

		Case 1	Case 2
Inputs	Symbols	13	10
	Resource Blocks	51	133
	MCS Table	64-QAM	256-QAM
	MCS Index	6	10
	Slots	1	2
Outputs	REs	7,650	15,162
	Total REs with Multiple Slots	7,650	30,324
	TBS	6,656	38,936
	Modulation Scheme	QPSK	16-QAM

Table 5

PHY layer processing stages.

IP Core	Case 1 (Bits)	Case 2 (Bits)
Signal Generator	6,656	38,936
CRC	7,040	42,240
SD-FEC	21,120	253,440
Bit Selection	15,300	121,280
Bit Interleaving and Modulation	7,650	30,320

Signal to Noise Ratio (SNR), where only a QPSK modulation scheme would be viable. The number of bits following each IP core can be confirmed in the Case 1 column of Table 5, which exactly match the expected results (derived from the equations and algorithms shown in Section 2.2).

The allocated REs found as Eq. (1b) whilst TBS can be found from Eqs. (2), (3b) and (4c), and confirmed through the Case 1 column of Table 4.

As a BG1 is selected and the TBS is smaller than 8448, only a single code block is required. As such, a 24-bit CRC is appended before padding with null bits based on the lifting size Z_c , which is selected from a predefined table of values. For BG1, the padded codeword can be found as $22 \cdot 320 = 7,040$, where $Z_c = 320$. The SD-FEC output size is also based on Z_c , for BG1 this can be found as $N = 66 \cdot 320 = 21,120$. A number of bits is selected from the SD-FEC output, containing data and parity bits, to fill the allocated resource elements with modulated data following Eq. (6).

The E_r bits are QPSK modulated following an interleaving process, producing $15,300/2 = 7,650$ REs, each containing 2 data bits through QPSK modulation. Thirteen symbols are allocated for data, with half of the subcarriers in one of these symbols reserved for DM-RS. This corresponds to 6 of the 12 subcarriers in each PRB in one symbol, therefore $DM - RS = 6 \cdot 51 = 306$ subcarriers. The PDSCH data perfectly fills the remaining allocated resources across the 13 symbols. Fig. 18 consists of user input/output and the two tabbed plots captured through the PYNQ interface for Case Study 1. Note the QPSK data

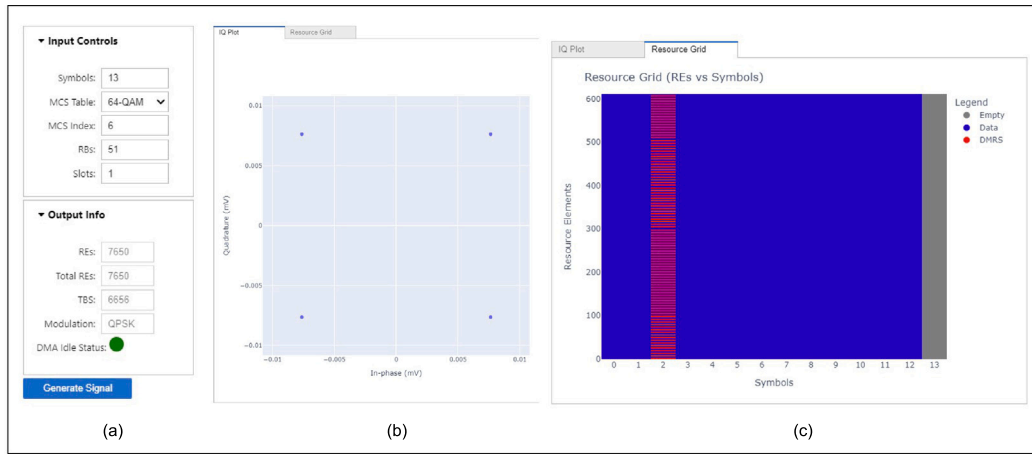


Fig. 18. PYNQ test harness GUI with (a) User input/output, (b) Constellation and (c) Resource grid plots for case study 1.

captured from the PDSCH pipeline. The resource grid plot comprises 13 symbols, each containing 51 PRBs (612 subcarriers). DM-RS can be seen on the third symbol, filling 306 subcarriers. The final symbol is empty and can either be allocated to additional channels, or left empty as a buffer between data symbols, to aid receiver synchronisation.

5.1.2. Case 2: 133 Allocated PRBs, 16-QAM modulation

For the second case, 133 PRBs are used, corresponding to a 50 MHz channel with 30 kHz subcarrier spacing. The user inputs and outputs from the PYNQ test harness can be seen in the Case 2 column of Table 4. Similarly, the output data lengths of each IP core can be seen as the Case 2 column of Table 5. Ten symbols are allocated to data, with MCS table values chosen to produce 16-QAM modulation.

Two slots are selected, and so the initial signal generation is repeated once. As in the previous case, REs can be found following (1b). In this instance, Eqs. (2), (3b) and (4b) are used to calculate TBS. As the TBS value is greater than 8,448, the generated data is segmented into code blocks based on (5). Each block is then padded to 8,448 prior to LDPC encoding.

Each code block is processed individually by the hardened SD-FEC, producing five LDPC encoded blocks, each of size 25,344. As two slots selected, 10 code blocks are produced following the SD-FEC, for a total of 253,440 bits. A number of bits are selected from each block as E_r . As there are multiple code blocks and the REs do not produce an integer when divided by the number of code blocks, Algorithm 1 is applied to determine the correct number of output bits to match the allocated REs. Here, three code blocks will be floored, $j = 0, 1, 2$ and two will take the ceiling value, $j = 3, 4$ for each slot. From these code blocks $N_{bits} = 3 \cdot 12,128 + 2 \cdot 12,132 = 60,648$ bits are extracted to perform 16-QAM modulation. From these bits an RE number of 16-QAM symbols can be produced, $N_{RE} = 60,648/4 = 15,162$.

As before, each second subcarrier in the third symbol is allocated to a DM-RS signal. The total DM-RS can be found as $DM - RS = 6 * 133 = 798$ subcarriers. The user input/outputs, 16-QAM constellation plot, as well as the resource grid of PDSCH data from the DMA can be seen in Fig. 19. Note that two slots-worth of PDSCH data has been produced with ten symbols per slot, leaving four empty symbols on each slot which may be allocated to other channels. The DM-RS symbol is produced in the same symbol on each slot and there are 133 PRBs (1,596 subcarriers) per symbol. To confirm that DM-RS occupies every second subcarrier in the appropriate symbol, the final plot within Fig. 19 presents the same resource grid but is focused on REs 1,430 to 1,457 of the third symbol. It can be observed that every even subcarrier is occupied by the red DM-RS signal.

5.1.3. Validation of design

For a 5G NR RU to be deployment viable, the radio frame must be generated and transmitted within the window of the Hybrid Automatic Repeat Request (HARQ), a PHY layer message which indicates the success or failure for a UE decoding PDSCH. 3GPP define a strict deadline for the HARQ signal, occurring between 1 and 15 slots after the PDSCH [57,65].

The inherent latencies of the PHY processing chain are variable based on the TBS and modulation scheme. Latency measurements have been recorded from TB generation through to the end of the presented PHY chain for the case studies in Sections 5.1.1 and 5.1.2.

For the first case, the transmitter blocks have a total latency of 40777 samples (0.332 ms at $f_s = 122.88$ MHz), and therefore can be generated in less than one slot. This latency leaves a margin of 0.168 ms for the remaining stages of PDSCH generation to meet the minimum HARQ deadline of one slot (0.5 ms with 30 kHz SCS).

For the second case, a total latency of 88303 samples has been measured (0.719 ms at $f_s = 122.88$ MHz). As such this larger case takes longer than 1 slot to generate, meaning the minimum HARQ timing cannot be met. However, in 5G NR the HARQ timing is not fixed and must be sent between 1 and 15 slots following PDSCH, so this larger case can be scheduled with a HARQ 2 or more slots after PDSCH.

Both cases can be seen to be deployment viable from the perspective of HARQ deadlines. Case 1 demonstrates that smaller TBS can be used to meet the minimum requirement of HARQ positioned a single slot after the PDSCH. Case 2 meanwhile showcases that although this design requires more than a single slot to generate a large PDSCH, this still meets the requirements, if HARQ is scheduled two or more slots following PDSCH. Notably, over 50% of the delay in both cases can be attributed to the bit interleaving process, where all bits are buffered in BRAM before being read out according to Algorithm 2.

6. Discussion

This paper has investigated an RFSoc device, alongside the PYNQ framework, as a single-chip Split-6 PHY 5G NR implementation platform. Three key contributions have been presented: an implementation of the PDSCH PHY layer using RFSoc, a single-chip Split-6 test harness and a demonstration of RFSoc Split-6 feasibility through hardware results and case studies.

The task of implementing the PDSCH on an RFSoc chip has been approached through the creation of a series of IP cores representing stages of the PDSCH PHY layer channel, alongside use of the SD-FEC hardened resources available on the RFSoc, to implement an RFSoc PHY RU. To the best of the authors knowledge this is the first published RFSoc implementation which performs all processing stages of PDSCH,

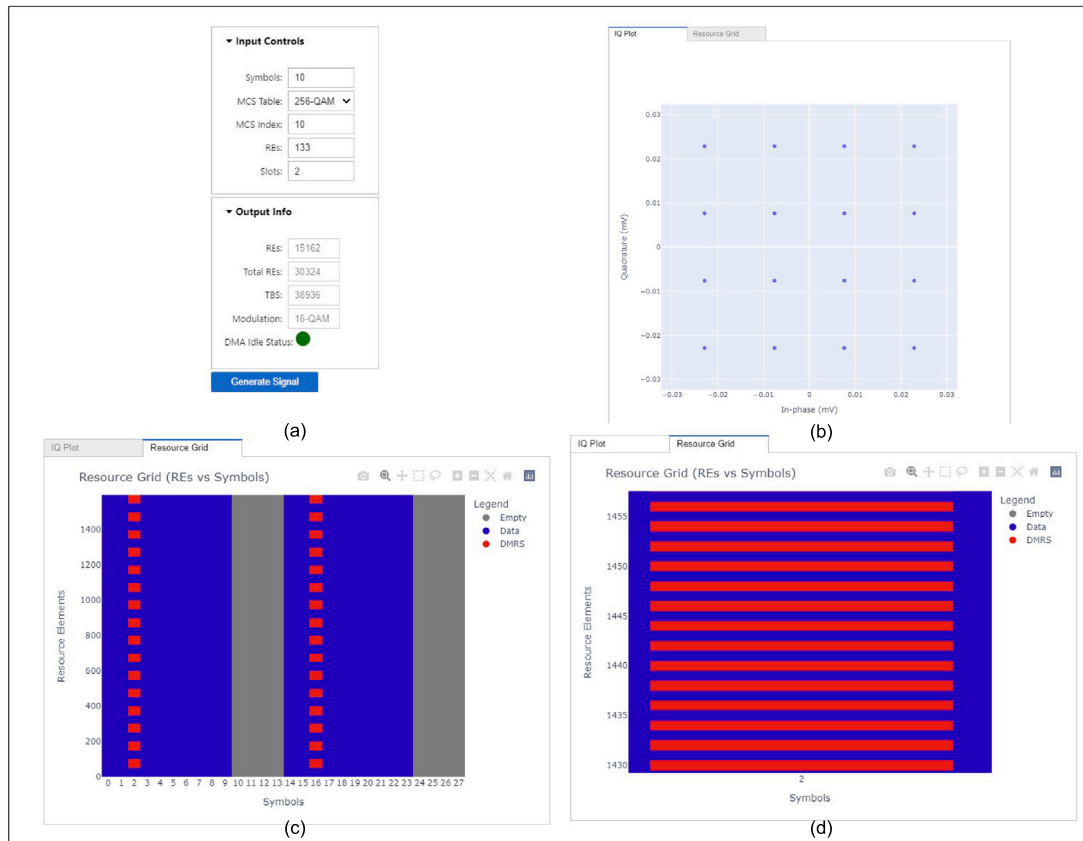


Fig. 19. PYNQ test harness GUI with (a) User input/output, (b) Constellation, (c) Resource grid and (d) DM-RS zoomed plots for case study 2.

starting with unprocessed data. The PDSCH processing is represented with four custom IP cores: *CRC and Code Block Segmentation*, *SD-FEC Control*, *Bit Selection* and *Bit Interleaving and QAM Modulation*.

An efficient implementation on the PL was presented in Section 4. The results show that the PDSCH design uses only a fraction of the available PL resources on the ZU48DR RFSoc. As such, other key PHY layer channels can be implemented alongside this design, namely: PDCCH, PRACH and PBCH, to create a full S-RU design. Also of note is the achieved clock speed of this design, meeting implementation timing at 122.88 MHz. This demonstrates the ability of the system to process data with 100 MHz bandwidths or lower through decimation, and as such the PDSCH implementation is suitable for any FR1 network.

Through the case studies in Section 5.1 the success of the custom IP cores can be assessed. As the calculations expressed here match the recorded results after each core, it can be validated that the RFSoc4x2 design has correctly implemented the PDSCH data processing steps.

Alongside the PDSCH implementation, a test harness has been created through additional custom IP cores and the PYNQ framework. On the ZU48DR RFSoc Chip PL a *Resource Allocation Controller* IP core takes user input through AXI4-Lite and automatically calculates RE allocation and transport block size. The values from this IP core update the PHY layer IP cores via AXI4-Lite. A *Signal Generator* IP core then produces data bits equal to the TBS. The *Resource Allocation Controller* IP core is updated from user input Python widgets through PYNQ. As these are memory-mapped interfaces, if one or parameters are changed via widget(s) the corresponding IP core is updated in real-time. The user widgets are paired with both QAM constellation and resource grid plots using processed data via DMA. The PYNQ test harness provides additional evidence of successful PHY layer processing, data bits have been selected from the LDPC output to produce the correct QAM modulation within each allocated resource element across the selected number of symbols. The PYNQ framework therefore enables a user

friendly approach to PDSCH testing without the need for CU/DU vRAN on a separate computing resource to the RU. Through the extensive background section, presentation of HDL Coder models and worked case study examples, this paper aims to serve as a guide to the transport block process found as [57] and the PDSCH data processing found as [56].

This study has focused on the implementation of the PDSCH PHY processing chain, presenting a practical evaluation of its deployment on the RFSoc PL. While control channels such as the PDCCH, PRACH and PBCH have not been included, and the PDSCH has not yet been processed into an OFDMA signal through an inverse Fast Fourier Transform, the focus on PHY-layer processing has allowed for a detailed investigation of how these critical functions can be realised in hardware. The results demonstrate that the RFSoc is a strong candidate for enabling Split-6 RU functionality.

The latency measurements presented in Section 5.1.3 further illustrate the behaviour of the design under different TBS and modulation schemes. In cases with large TBS values and higher-order modulation, processing extends beyond a single slot, meaning the minimum HARQ timing of one slot after the PDSCH is not achieved in all instances. The measured end-to-end latency in both case studies show that HARQ deadlines remain within reach of the system. These findings suggest that, with the integration of additional PHY-layer channels, the RFSoc-based design is a viable platform for full Split-6 RU realisation.

7. Future work

A full implementation of the PDSCH PHY layer has been presented through this work combined with user control and visualisation of a resource grid for one or more slots of data. The future developments of this project are therefore clear; implementing the key remaining physical channels of 5G NR within the FPGA PL to produce a signal

which can be synchronised to a UE. Of note this should include PDCCH, PRACH and PBCH. These additional should be integrated into the software control element of the system, alongside real data generation, to create a complete Split-6 RU platform based on the RFSoc.

At present, validation is limited to signal generation and PHY-layer processes driven by test signals generated on the PS of the same chip. This purposefully creates a single chip test and validation platform for a Split-6 RU. There is clear scope to extend the presented work to include a live MAC layer alongside TBs of raw data from an external source, provided to the RFSoc over nFAP or a custom UDP Ethernet packet. This is a natural next step to ensure that the precise timing required for a Split-6 network can be met with the presented PHY-layer implementation.

8. Conclusion

This paper has confirmed the feasibility of RFSoc devices as candidate small cell Split-6 PHY RUs. The key contributions have been achieved through the implementation of PDSCH PHY layer processing implemented in the PL of a ZU48DR RFSoc using an RFSoc4x2 board, via a series of custom IP cores and the hardened SD-FEC. Software based on the PYNQ framework has been implemented as a low-complexity test harness alongside the PDSCH channel which enables the arbitrary allocation of PDSCH data within a radio slot, produced through the hardware pipeline and visualised to the user. The functionality of the PDSCH implementation has been validated with live testing on an RFSoc4x2 board through two case studies.

The hardware resource usage of the design, both with and without the test harness, has been evaluated. Results generated from Vivado show that the design uses only a fraction of the available PL resources in both cases. The RFSoc PHY RU design uses less than 10% of BRAM and under 3% of other PL resources. This allows further implementation of PHY layer channels, and meets timing at 122.88 MHz, which enables 100 MHz bandwidth channels.

The PDSCH channel has been validated with two test cases, simulating the resource allocation of a 20 MHz and a 50 MHz cell. This method produces real, 3GPP standardised PHY data encompassed in a single platform with S-DU control information, providing an accessible data plane. The presented Split-6 RFSoc design can be taken further by including appropriate control channels, etc. to create a full S-RU, which could be combined with the hardened RF Data Converters of the RFSoc to produce transmittable 5G NR signals using a single platform.

The findings of this work establish the RFSoc chip as a practical hardware platform for Split-6 RU implementation, moving beyond simulation-based feasibility studies to a functioning prototype and thus helping to bridge the gap between theoretical investigations and deployment-ready solutions. The demonstration of the RFSoc chip as a viable Split-6 RU presented in this paper highlights the potential of RFSoc to support low-power, small-cell architectures. This work therefore provides a foundation for future research focused on optimisation, interoperability with MAC layer protocols, and large-scale experimental validation. In this way, the work contributes to advancing Split-6 as a viable approach for real-world 5G NR deployments.

CRedit authorship contribution statement

Lewis J. Brown: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Louise H. Crockett:** Writing – review & editing, Supervision, Resources. **Robert W. Stewart:** Supervision, Project administration.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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